

陶加焱

网页 UI 设计师 | 物理学家 | Agent 应用开发者

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PROFESSIONAL SUMMARY

Web UI Design: 独立设计并开发响应式个人学术主页 (tao-jiayan.pages.dev)，采用 Swiss Modernism 极简风格，兼容桌面端与移动端，通过 Cloudflare Pages、GitHub Pages 与 Gitee 三线部署，具备完整的 SEO 优化与 JSON-LD 结构化数据标记。

Computational Physics: 广州大学物理学本硕连读 (GPA 3.76/4.0)，运用非平衡格林函数 (NEGF) 方法模拟介观量子输运，独立撰写两篇凝聚态物理论文，所有数值计算程序开源托管于 github.com/jiatao058-afk/my-code。

Agent Application Development: 深度掌握 AI Agent 开发工具链 (Claude Code)，熟练运用大语言模型进行软件架构设计、自动化脚本编写及代码调试，将 AI 工具落地为真实工程生产力。

AREAS OF EXPERTISE

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|--|---|---|--|
| ☐ Python | ☐ NumPy/SciPy/Matplotlib | ☐ Web UI Design | ☐ HTML/CSS/JavaScript |
| ☐ Playwright Web Scraping | ☐ Git Version Control | ☐ NEGF Quantum Transport | ☐ LaTeX |
| ☐ Agent Applications | ☐ Claude Code | ☐ MATLAB | ☐ Responsive Design |

PROFESSIONAL EXPERIENCE

Guangzhou University — Department of Physics

Graduate Research Assistant | 2025 - Present | Guangzhou, CN

- Built NEGF-based numerical framework to simulate electron transport in mesoscopic devices, modeling Hamiltonian matrices, self-energy, and transmission coefficients with Python (NumPy/SciPy/Kwant)
- Studied dephasing effects in zigzag graphene nanotubes using Buttiker virtual probe formalism, analyzing conductance dependence on device length, temperature, and dephasing strength
- Independently completed two condensed matter physics manuscripts from theoretical modeling through numerical solution to final manuscript preparation in LaTeX
- Published all computational code as open-source at github.com/jiatao058-afk/my-code for reproducibility

Independent Projects — Web & Automation

Full-Stack Developer | 2025 - Present | Remote

- Designed and built a responsive personal academic homepage (tao-jiayan.pages.dev) with SEO-optimized semantic HTML5, JSON-LD structured data, and CSS3 animations; deployed across Cloudflare Pages, GitHub Pages, and Gitee Pages
- Developed browser automation tools using Playwright framework for web data collection, automated form filling, and batch workflow processing
- Built interactive browser games (rock-paper-scissors, reaction test) using Vue 3 + TypeScript, covering full development cycle from UI design to production deployment

CERTIFICATIONS